

Table 1: CPU time on dataset collection 3 with 15.0% constraints

| Constraints Level<br>Method<br>Dataset | 0.15    |       |         |         |         |       |
|----------------------------------------|---------|-------|---------|---------|---------|-------|
|                                        | COPKM_P | KSKM  | KSKM_E  | BLPKMCC | PCCC    | COPKM |
| synthetic_n1000_d2_k10                 | 0.01    | 0.03  | 0.17    | 3.01    | 1.01    | 0.01  |
| synthetic_n1000_d2_k100                | 0.06    | 0.87  | 230.95  | 153.35  | 388.15  | 0.12  |
| synthetic_n1000_d2_k2                  | 0.00    | 0.00  | 0.00    | 0.29    | 0.01    | 0.00  |
| synthetic_n1000_d2_k20                 | 0.02    | 0.14  | 10.26   | 24.53   | 24.67   | -     |
| synthetic_n1000_d2_k5                  | 0.00    | 0.01  | 0.01    | 0.90    | 0.06    | 0.00  |
| synthetic_n1000_d2_k50                 | 0.04    | 0.65  | 145.85  | 92.83   | 199.81  | 0.08  |
| synthetic_n2000_d2_k10                 | 0.01    | 0.02  | 0.07    | 8.68    | 0.68    | 0.01  |
| synthetic_n2000_d2_k100                | 0.15    | 4.08  | 1650.63 | 1146.53 | 2694.62 | 0.26  |
| synthetic_n2000_d2_k2                  | 0.01    | 0.01  | 0.01    | 1.14    | 0.05    | 0.01  |
| synthetic_n2000_d2_k20                 | 0.02    | 0.10  | 3.33    | 29.45   | 11.27   | 0.03  |
| synthetic_n2000_d2_k5                  | 0.01    | 0.01  | 0.01    | 2.91    | 0.06    | 0.01  |
| synthetic_n2000_d2_k50                 | 0.08    | 1.21  | 1304.08 | 598.16  | 776.39  | -     |
| synthetic_n5000_d2_k10                 | 0.05    | 0.04  | 0.05    | 38.34   | 0.32    | 0.05  |
| synthetic_n5000_d2_k100                | 0.35    | 13.13 | 3611.41 | 3763.40 | 3700.85 | -     |
| synthetic_n5000_d2_k2                  | 0.04    | 0.03  | 0.03    | 7.22    | 0.32    | 0.05  |
| synthetic_n5000_d2_k20                 | 0.06    | 0.06  | 0.33    | 121.39  | 4.17    | 0.06  |
| synthetic_n5000_d2_k5                  | 0.05    | 0.04  | 0.04    | 18.34   | 0.29    | 0.06  |
| synthetic_n5000_d2_k50                 | 0.11    | 0.52  | 1038.47 | 641.14  | 195.52  | -     |
| synthetic_n500_d2_k10                  | 0.01    | 0.02  | 0.12    | 1.48    | 1.72    | 0.01  |
| synthetic_n500_d2_k100                 | 0.03    | 0.33  | 69.36   | 29.83   | 63.97   | 0.05  |
| synthetic_n500_d2_k2                   | 0.00    | 0.00  | 0.00    | 0.09    | 0.02    | 0.00  |
| synthetic_n500_d2_k20                  | 0.01    | 0.22  | 7.66    | 6.24    | 14.43   | 0.02  |
| synthetic_n500_d2_k5                   | 0.00    | 0.01  | 0.01    | 0.31    | 0.13    | 0.00  |
| synthetic_n500_d2_k50                  | 0.02    | 0.38  | 27.45   | 14.92   | 35.62   | 0.03  |